

1. Data Quality & Suitability

1.1 Accuracy and fit-for-use

What it means: The data used to develop PD/LGD should correctly represent the underlying mortgage, borrower, default, collateral, claim, recovery, and loss information. “Fit-for-use” additionally means that even technically correct data must be appropriate for the model's purpose.

What you validate: Start with basic profiling: record counts, duplicates, missing values, invalid values, unusual zeros, extreme observations, inconsistent categories, and impossible combinations. Then perform business-rule checks. For example, origination should generally precede observation/default, default should precede subsequent claim/recovery events, property values and balances should be plausible, and calculated variables such as LTV should reconcile.

For important derived variables, independently recalculate them. If LTV is defined as mortgage balance divided by property value, reproduce it. For LGD, independently reproduce EAD, recoveries, expenses, discounting and realized loss for a sample of defaults.

PD examples: Validate default flags, default dates, credit score, LTV, loan balance, delinquency history, amortization, property information and other major risk drivers.

LGD examples: Validate EAD, claim amounts, property sale proceeds, recoveries, legal/foreclosure expenses, recovery dates, workout completion and realized LGD.

Your conclusion: Are errors sufficiently small and controlled that they are unlikely to materially affect model development, calibration, performance or conclusions?

1.2 Relevance and representativeness

There are two different questions here.

Relevance: Is this the right information for the model's intended purpose? A variable could be perfectly accurate but irrelevant to mortgage default/loss modelling.

Representativeness: Does the development sample adequately represent the population on which the model will actually operate?

Compare development data against the intended/current portfolio across important dimensions such as:

- origination vintage;
- geography/province;
- LTV;
- credit score;
- loan balance;
- property type;
- lender/channel;
- product type;
- amortization;
- borrower characteristics;
- default rates;
- economic environment.

For LGD, additionally compare characteristics of the default/workout sample, including EAD, LTV, recovery rates, foreclosure rates, workout duration and loss severity.

You should also examine **economic representativeness**. If development data come primarily from benign housing conditions, ask whether they adequately represent periods of falling house prices, elevated unemployment or other relevant stresses.

A particularly useful analysis is:

Development population vs validation population vs current production population.

Large differences aren't automatically problems. You need to determine whether they undermine model applicability.

1.3 Data lineage and provenance

Lineage asks where the data travelled.

Provenance asks where the data originally came from and what happened to it.

For important variables, you should be able to trace:

Original source → extraction → intermediate tables → cleansing → transformation → modelling dataset → model feature

For example:

Property valuation system → raw property value → HPI adjustment → estimated current property value → calculated LTV → LGD input

Check source-system names, source fields, extraction logic, joins, filters, transformations, aggregation rules and final model fields.

Independently trace a sample of records end-to-end.

Pay special attention to externally sourced information such as credit bureau information, property valuations, house-price indices and macroeconomic variables.

The question you ultimately want answered is:

Can another qualified person determine exactly where every material model input came from and reproduce it?

1.4 Timeliness

Correct information can still be unsuitable if it is stale.

Determine:

- when data were observed;
- when data were extracted;
- how frequently they are updated;
- how much reporting lag exists;
- whether stale records exist;
- how stale records are handled;
- whether development and production use comparable timing conventions.

This can be particularly important for LGD. Suppose a property value used to calculate current LTV is substantially outdated during a rapidly changing housing market. The data may have been accurate at the valuation date but inappropriate at the prediction date.

For important dynamic variables, calculate:

Model reference/scoring date – input observation date

Then examine the distribution.

1.5 Privacy, regulatory, ethical and internal compliance

You don't necessarily perform the institution's legal/privacy assessment yourself. As validator, however, you should establish that appropriate governance and approvals exist.

Check whether:

- data use is authorized;
- customer information is used for an approved purpose;
- privacy requirements are addressed;
- access is appropriately controlled;
- sensitive variables are appropriately governed;
- internal data policies are followed;
- third-party data restrictions are respected;
- retention requirements are followed;
- ethical/fairness requirements are considered.

Pay particular attention to variables that might act as **proxies** for sensitive characteristics—for example, highly granular geographic information.

Your validation evidence might therefore include privacy/governance approvals rather than an independent legal conclusion.

1.6 Synthetic and proxy data

These are different issues.

Synthetic data are artificially generated observations/data.

If used, determine why they were needed, how they were generated, what proportion of development data they represent, which variables are synthetic and whether the generated data preserve realistic distributions and relationships.

For PD/LGD, be particularly cautious if synthetic observations are being used to represent **rare defaults or severe losses**. Synthetic observations should not create artificial model performance or unrealistic tail behaviour.

Proxy data are substitutes for information that isn't directly available.

For example:

Actual current property value unavailable → HPI-adjusted property value used as proxy.

Assess whether the proxy reasonably represents the underlying concept. Compare proxy values against actual observations wherever possible and determine whether errors vary systematically by geography, property type, vintage or economic environment.

1.7 Data cleansing

Developers will rarely model completely raw data. You need to understand every material cleansing operation.

Examples include:

- removing duplicates;
- excluding invalid records;
- correcting errors;
- imputing missing values;
- capping/flooring;
- winsorizing;
- combining categories;
- removing outliers.

For every material cleansing rule ask:

Why was this necessary?

How many observations were affected?

Is the rule reasonable?

Could it remove genuine risk information?

How sensitive is the model to the treatment?

For example, don't automatically accept removing extremely high LGDs as “outliers.” They might represent exactly the severe mortgage losses that an LGD model needs to understand.

1.8 Development, validation and production consistency

This has two dimensions.

First, ensure the **independent validation dataset** is constructed consistently enough that comparisons with development results are meaningful.

Second, ensure eventual **production data definitions and transformations** are consistent with what the model was developed on.

Compare:

Development → Validation → Production

for source fields, definitions, units, missing-value handling, categorical mappings, transformations, timing conventions and calculated variables.

A classic problem is:

Development defines LTV using one property-value source while production calculates it from another.

Even though the formula is identical, the model isn't receiving equivalent information.

1.9 Data exclusions and bias

Every material exclusion should be challenged.

Examples:

- mortgages missing credit scores removed;
- particular lenders excluded;
- certain vintages removed;
- unusual property types excluded;
- incomplete LGD workouts removed.

Ask whether exclusions systematically remove certain kinds of observations.

Important forms include **selection bias**, where the development sample differs systematically from the target population; **survivorship bias**, where only observations that remain observable are included; and **censoring bias**, especially important for LGD where recent defaults may not yet have completed their workout process.

For LGD, compare completed and incomplete workouts. If incomplete cases systematically have higher LTV, different geography, longer foreclosure periods, or other characteristics, using only completed workouts may bias the model.

1.10 Material impact of data-quality problems

Not every data issue should become a major validation finding.

You need to determine **materiality**.

For an identified problem, quantify where possible:

Data problem → affected observations → affected features → affected predictions → affected model performance/business use

For example, suppose 8% of property values are stale. Recalculate affected LTVs using better information where available and determine whether LGD predictions materially change.

This distinguishes a technical data-quality observation from an actual **model-risk issue**.